

● MEDIUM

● ALGEBRA

Linear equations in one variable

01

10 questions · ~15 min

Q1

ALGEBRA

If $46 = 16 + 2x - 8$, what is the value of $2x - 8$?

A. 16

B. 23

C. 30

D. 38

$$(b - 2)x = 8$$

In the given equation, b is a constant. If the equation has no solution, what is the value of b ?

- A. 2
- B. 4
- C. 6
- D. 10

Q3

GRID-IN ALGEBRA

$$2x + 16 = ax + 8$$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A bowl contains 20 ounces of water. When the bowl is uncovered, the amount of water in the bowl decreases by 1 ounce every 4 days. If 9 ounces of water remain in this bowl, for how many days has it been uncovered?

- A. 3
- B. 7
- C. 36
- D. 44

Megan's regular wage at her job is p dollars per hour for the first 8 hours of work in a day plus 1.5 times her regular hourly wage for work in excess of 8 hours that day. On a given day, Megan worked for 10 hours, and her total earnings for that day were \$137.50. What is Megan's regular hourly wage?

- A. \$11.75
- B. \$12.50
- C. \$13.25
- D. \$13.75

$$4x + 12 = \frac{ax + 3}{2}$$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a ?

- A. 0
- B. 3
- C. 8
- D. 12

If $2(x-5)+3(x-5)=10$, what is the value of $x-5$?

- A. 2
- B. 5
- C. 7
- D. 12

Q8

GRID-IN

ALGEBRA

$$2(p + 1) + 8(p - 1) = 5p$$

What value of p is the solution of the equation above?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$66x = 66x$$

How many solutions does the given equation have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

The cost to rent a bus from Company X is \$ 950 for the first 3 hours and an additional \$ 50 per hour for each hour after the first 3 hours. If the total cost to rent the bus from Company X for t hours, where $t > 3$, is \$ 1,150, which equation represents this situation?

A. $950(t - 3) + 50t = 1,150$

B. $950(3t) + 50t = 1,150$

C. $950 + 50(t - 3) = 1,150$

D. $950 + 50(3t) = 1,150$